

Comrade Business One AI TOOL

Astran Auto-Repair System

User & Installation Manual | Version 1.0.0

Detects missing astran postings before they cause financial loss

Repairs invoices automatically - Sales / VAT / Customer / Cost / Stock

Live dashboard with branch-wise status and one-click repair

Comrade Business One AI TOOL v1.0.0

Astran Auto-Repair System - User & Installation Manual

1. What This Tool Does

The Comrade Business One AI TOOL monitors your **astran** posting table and prevents financial loss caused by **missing sales-postings**.

When invoices (SINVOICE) are created in **Comrade Business One**, the corresponding posting entries must be written to the astran table (sales, VAT, customer, cost and stock rows). If these postings fail or are skipped, the invoice is "posted in SAP but missing in astran" - which causes reports, VAT returns and customer balances to be wrong.

This tool:

- **Scans** every branch's invoices for the selected period.
- **Detects** any invoice that is missing its astran postings.
- **Repairs automatically** - it re-creates the 5 required astran rows

(SALES, VAT, CUSTOMER, COST, STOCK) using the correct accounts from assys and the actual unit cost from asicm/asict.

- **Reports** the results on screen and (optionally) by email.

The tool runs as a local web application on the client PC. No SAP license or extra software is needed on the client machine beyond the ODBC driver (see Section 3).

2. Contents of the Package

The installation package is a single ZIP file:

File / Folder	Purpose
ComradeAITool.exe	Main application (starts the dashboard on port 8503).
StartComradeAITool.bat	Double-click helper that starts the app and opens your browser.
RunComradeAITool.bat	Manual server run with watchdog - auto-restarts if the server is cut off.
_internal\	Program files (Streamlit runtime + hidden business engine).
astran_config.json	Settings file (server, database, branches, email, schedule).
astran_repair.log	Activity log written automatically by the tool.
.streamlit\	Web server settings (port 8503). Created automatically.

Business logic protects your data: the core engine (astran_repair.py) is compiled into the application and is **not** shipped as readable source, so the posting rules cannot be tampered with.

3. System Requirements (Client PC)

Requirement	Minimum
Operating system	Windows 10 / 11 (64-bit), Server 2016+
RAM	4 GB (8 GB recommended)
Disk space	300 MB free
Database access	Network access to the Comrade Business One SQL Server
ODBC Driver	Microsoft ODBC Driver 17 for SQL Server (must be installed, see below)
Browser	Chrome, Edge or Firefox (any recent version)
Python	Not required - bundled in the app

3.1 Install the ODBC Driver 17

The tool connects to SQL Server through the **ODBC Driver 17 for SQL Server**. This is the single most common cause of "Driver not found" errors, so install it first.

1. Download `msodbcsql17_x64.msi` from the Microsoft Download Center, or use the ODBC installer supplied with your deployment.
2. Run it and choose **Complete** installation.
3. Verify: open `C:\Windows\System32\odbcad32.exe` -> **Drivers** tab ->

you should see "**ODBC Driver 17 for SQL Server**" listed.

Note: if the target machine is 32-bit only, the tool will not run - use a 64-bit machine.

4. Installation at the Client Site

Follow these steps in order on the client PC:

1. **Install the ODBC Driver 17** (Section 3.1) if not already present.
2. Copy `ComradeAITool.zip` to the client PC (e.g. to Downloads).
3. **Extract the ZIP:**
 - Right-click `ComradeAITool.zip` -> **Extract All...**
 - Choose a permanent folder, e.g. `C:\ComradeTool` (avoid Desktop/Downloads, which can be cleaned by some policies).
 - The extracted folder must contain `ComradeAITool.exe` and `StartComradeAITool.bat` **directly** (do not extract to a sub-folder inside the ZIP's own folder name).
4. (First run only) **Configure the database** - either:
 - **Option A - edit the settings file:** open `astran_config.json` in Notepad and enter the client's server, database and credentials (see Section 5), **or**

- **Option B - use the Settings page:** start the tool, go to **Settings**, enter server/database/username/password and click **Test Connection** and then

Save Settings (Section 6.4).

5. **Start the tool** (Section 4.1).

4.1 Starting the Tool

- **Double-click** `StartComradeAITool.bat`. A command window opens showing the startup messages, and your browser opens automatically at `http://localhost:8503`.

- Alternatively double-click `ComradeAITool.exe`.

- **Keep the command window open** while using the tool. Closing it stops the tool.

(For unattended operation see Section 8.)

4.2 Running the Server Manually with Watchdog

If the server sometimes **gets cut off / stops on its own**, use `RunComradeAITool.bat` instead of the normal start file:

1. **Double-click** `RunComradeAITool.bat`.
2. A window titled `"Comrade AI TOOL - Server (Watchdog Active)"` opens and starts the server live in that window.
3. If the server stops or crashes for any reason, the window **restarts it automatically** after 8 seconds and continues forever, until you close it.
4. To stop it permanently, press **Ctrl+C** in that window (and answer the "Terminate batch job?" prompt with **Y**).

This is the recommended launcher on a PC where the server must keep running all day.

If port 8503 is already in use, the tool detects it and just opens the browser to the already-running instance.

4.2 Stopping the Tool

- Close the command window, **or**
- Press `Ctrl+C` in the window, **or**
- Run: `taskkill /f /im ComradeAITool.exe`

5. The Settings File (`astran_config.json`)

This file sits **next to** `ComradeAITool.exe` and holds all configuration. It is read at startup and updated by the Settings page.

```
{
  "server": "192.168.1.1",
  "database": "comrade",
  "username": "admin",
  "password": "123456",
  "branches": ["HO", "SH"],
  "email": {
    "smtp_server": "smtp.gmail.com",
```

```

"smtp_port": 587,
"from": [REDACTED],
"to": [REDACTED],
"password": [REDACTED],
},
"schedule": {
"enabled": true,
"interval_minutes": 30,
"auto_fix": true
}
}

```

Key	Meaning
server	SQL Server instance, e.g. SERVERNAME\INSTANCE or host,port.
database	Comrade Business One company database (e.g. [REDACTED]).
username / password	SQL login (e.g. sa).
branches	Branch codes to scan. Note: the tool loads branches from the <code>asdept</code> table automatically; this list is the fallback if the database cannot be reached.
email	SMTP settings for alert emails (Gmail example shown).
schedule	Auto-repair preferences (stored for automation, see Section 8).

5.1 Important notes for safe deployment

- The file is **readable and editable** - keep it in a folder only administrators can access, because it contains the SQL password in plain text.
- **Back it up** after configuration is complete, so a reinstall can restore settings in seconds.
- If `astran_config.json` is missing at startup, the tool creates an empty default file automatically (no credentials). You then configure it via the Settings page.

6. Using the Dashboard

The dashboard has four pages, chosen from the sidebar menu:

Page	Purpose
Dashboard	Overall picture: totals, missing counts, branch status, comparison chart.
Check & Repair	Scan for missing postings; repair everything in one click.
History	Show and clear the repair log.
Settings	Configure database, branches, email and schedule.

The header on every page shows the **connected database** and the **list of branches** currently being monitored.

6.1 Dashboard

- **Date Range** - choose **Current Year** (default) or **Custom Range** (pick From/To dates). Every number and chart on the page respects this range.

- **Statistic cards:**

- **Total Invoices** - all SINVOICE invoices in the selected range across branches.
- **Missing** - invoices that have no astran posting (need repair).
- **Fixed Today** - how many repairs were done today (from the log).
- **Branches** - number of branches monitored.
- **Branch Status** - a card per branch showing green **ALL OK** or amber

N MISSING.

- **Branch Comparison** - bar chart of missing invoices per branch (green = OK, amber = issues).

6.2 Check & Repair

This is the main working page.

1. **Date Range** - select the invoice period to scan (Current Year or Custom Range).
2. **Select Branches** - tick the branches to include (all are selected by default).
3. Then choose one of:

- **Check Now** - only scans and lists the missing invoices. **Nothing is written to the database.**

- **Fix All & Check** - repairs every missing invoice found (creates the astran rows), then re-scans to confirm, and sends an email report if email is configured.

4. Results are shown per branch:

- Amber warning card: "N missing invoice(s) found", listing each invoice number, date, customer and amount.

- Green card: "All records intact".

5. Use **Clear Results** to reset the view.

6.3 History

- Shows the last 50 log entries from `astran_repair.log` with colour coding: green = FIXED, blue = OK, red = ERROR, amber = other.

- **Clear History** deletes the log file (a new one is created on the next activity).

6.4 Settings

All settings are saved to `astran_config.json` when you click **Save Settings**.

- **Database Connection** - Server, Database, Username, Password.

- **Branches** - shown as "Detected branches", loaded from `asdept.deptno` once the connection works. You can override with the text box (one code per line).
- **Email Alerts** - SMTP server/port, From address, To addresses (one per line), and password.
- **Auto Repair Schedule** - Enable/disable, interval in minutes, and auto-fix toggle.

Buttons:

- **Save Settings** - stores everything. The form is cleared on success.
- **Test Connection** - verifies the SQL connection with the values shown (before saving).
- **Test Email** - sends a test email using the current email settings.

Use **Test Connection** first, then **Save** - never leave the page without saving after changing values.

7. Repair Logic Summary (what "Fix All" does)

For every missing invoice the tool creates **5 rows** in `astran`:

Row	jvtype	type	Amount used	Account source	costc
Sales	IS	C	Net amount (total - VAT)	<code>asssys.gl_cashsal</code>	SALES
VAT	IS	C	VAT amount	<code>asssys.GL_VAT_SALES</code>	VAT
Customer	IS	D	Invoice total	invoice customer	CUSTOMER
Cost	IS	D	Actual cost (asicm.cost_avg via asict)	<code>asssys.gl_cost</code>	COST
Stock	IS	C	Actual cost	<code>asssys.gl_stock</code>	STOCK

Key behaviour - **read this carefully for your auditors**:

- The tool **never deletes or modifies existing rows** - it only inserts missing ones.

Re-running **Fix All** is safe: invoices already posted are skipped.

- A row is considered "present" when an `astran` entry exists with the same `type_no` (invoice number), `jvtype` = 'IS' and `deptno`.
- If `asssys` has no account row for a branch, that branch is reported as **ERROR** and **not repaired** (nothing is written).
- All repairs for a branch are committed as one transaction; if anything fails, the whole branch is rolled back - no partial or double postings.

8. Making the Tool Run Automatically (optional, recommended)

Goal: ensure postings are always repaired even if nobody opens the dashboard, so no astran loss silently accumulates.

Option A - Auto-start at logon (easiest)

1. Press Win + R, type shell:startup, press Enter.
2. Right-click inside the folder -> **New -> Shortcut**.
3. Browse to the extracted folder and select StartComradeAITool.bat.
4. Name it "Comrade AI TOOL" and click Finish.

The tool now starts (and repairs can be run) whenever a user logs in.

Option B - Scheduled task at a set time (recommended)

Use **Task Scheduler** to open the dashboard automatically, e.g. every morning:

1. Open **Task Scheduler -> Create Basic Task...**
2. Name: ComradeAITool.
3. Trigger: **Daily**, at your preferred time (e.g. 8:00 AM).
4. Action: **Start a program** -> Program: browse to StartComradeAITool.bat

(working directory must be the extracted folder).

5. Finish. Set the machine to be powered on at that time.

Note on the stored schedule (interval): the Settings page saves an interval

(default 30 minutes) as configuration. Full unattended looping repairs are

planned as an enhancement; today the practical automation is the scheduled/startup

launch above plus the one-click **Fix All & Check**.

9. Email Alerts

Email alerts must be configured (Section 6.4) and use an SMTP server that accepts this app's logins:

- **Gmail (smtp.gmail.com:587)** - the "password" must be an **App Password**

(create at Google Account -> Security -> App passwords). A normal account password will be rejected unless 2-step verification + app password is set up.

- Use **Test Email** from Settings to confirm delivery before relying on it.
- The tool sends a **Repair Report** email after a **Fix All & Check** run, listing

branch, invoice number, date, customer, amount and rows inserted.

- If nothing was fixed, no email is sent (no spam).

10. Troubleshooting

Symptom	Cause / Fix
"Driver not found" / connection error	ODBC Driver 17 not installed - install it (Section 3.1).
Cannot connect / login failed	Wrong server, database, username or password in Settings or config file.

Symptom	Cause / Fix
No branches shown in Settings	Connection not saved/working - click Test Connection , then Save .
Tool won't start, port in use	Another instance running - the tool opens the browser to the existing one.
Dashboard one-time error after edit	Clear <code>__pycache__</code> and restart the tool.
Email not sending	Use an app password for Gmail; verify with Test Email ; check dept firewall for outbound port 587.
Branch shows ERROR, not repaired	assys has no account record for that branch code - add it in SAP, then re-run.
Port 8503 blocked after Windows update	Check firewall allows localhost; restart the tool.
Server keeps stopping / getting cut off	Use <code>RunComradeAITool.bat</code> - its watchdog restarts the server automatically. If the server still stops, check <code>astran_repair.log</code> and <code>launcher_error.log</code> and contact Comrade support.

Where to find logs

- `astran_repair.log` - next to `ComradeAITool.exe`. Every scan/repair is logged with a timestamp (FIXED / OK / ERROR lines).
- If the launcher itself fails, `launcher_error.log` appears next to the exe.

11. Checklist for a Successful Go-Live

- [] ODBC Driver 17 installed (verify in `odbcad32.exe`).
- [] ZIP extracted to a permanent folder (not Desktop/Downloads).
- [] `astran_config.json` configured OR Settings page saved successfully.
- [] **Test Connection** = Connected.
- [] **Test Email** = delivered (if email alerts needed).
- [] Dashboard header shows the correct database and all expected branches.
- [] Ran **Check Now** first (no writes) to see the current missing count.
- [] Ran **Fix All & Check** - count drops to 0 per branch, green status shown.
- [] Added shortcut to Startup or a scheduled task (Section 8).
- [] Backed up `astran_config.json`.
- [] Directed the client contact to this manual; logged the go-live date & baseline missing count.

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